UNIT – III

Boundary Layer Theory & Dimensional Analysis

Subject : Fluid Mechanics & Hydraulic Machines (23A03402T)

Syllabus: Boundary layer – introduction, momentum integral equation, displacement / momentum / energy thickness, separation & its control, streamlined & bluff bodies, velocity profiles. Dimensional analysis – homogeneity, non-dimensionalisation of equations, repeating-variable method, Buckingham π-theorem.

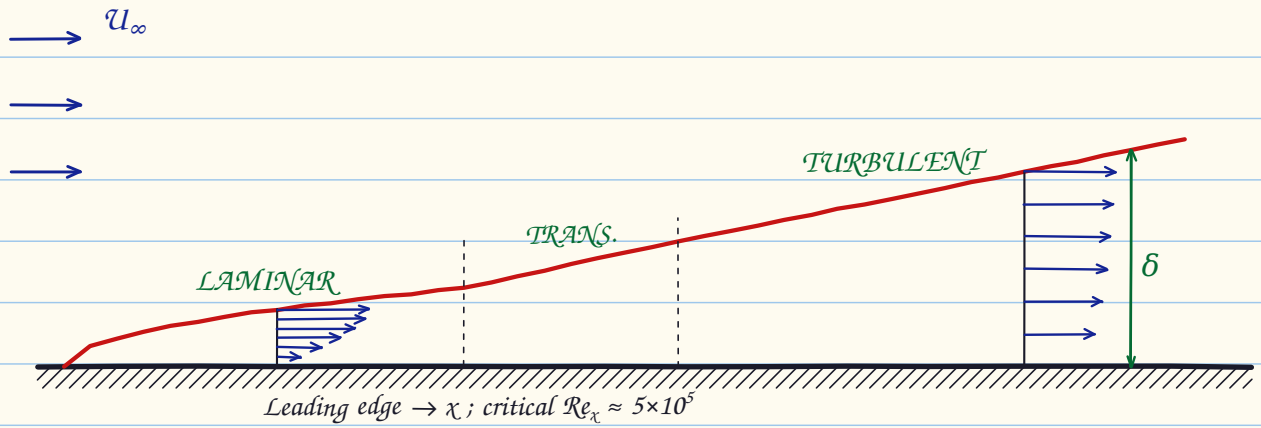
1. Boundary Layer Theory – Introduction

Ideal-fluid (potential) theory ignores viscosity and predicts zero drag (d'Alembert's paradox). Real fluids stick to a solid surface (no-slip), so velocity rises from zero at the wall to the free-stream value U_∞. Prandtl (1904) : the effect of viscosity is confined to a thin layer next to the surface – the boundary layer.

- Inside the layer : large velocity gradient $du/dy \Rightarrow$ significant shear stress $\tau = \mu\, du/dy$; viscous & inertia forces are of the same order.
- Outside the layer : fluid behaves as ideal (inviscid, irrotational) – potential flow, Bernoulli's equation can be used.
- **Boundary layer thickness δ** : distance from the wall at which $u = 0.99\, U_\infty$ (nominal thickness).

Growth of boundary layer on a flat plate

δ = thickness where $u = 0.99\ U_\infty$



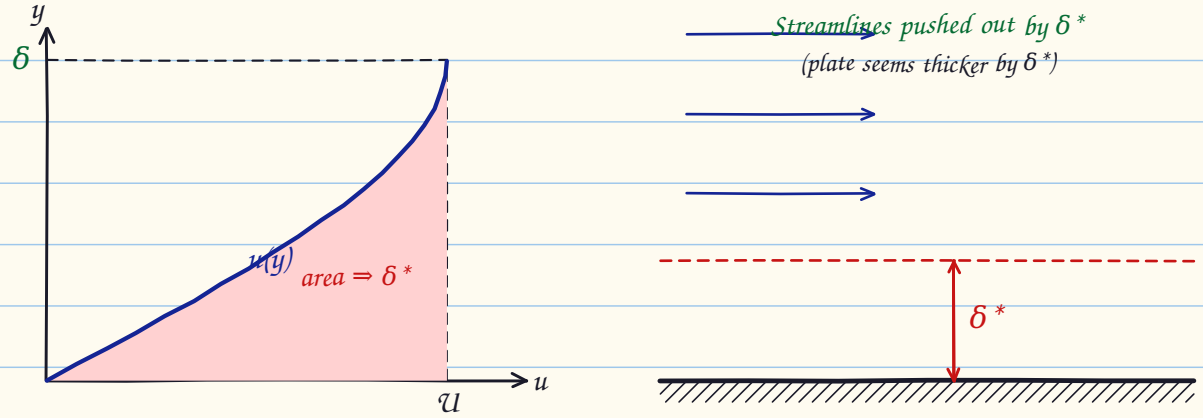
- Near the leading edge the layer is laminar (δ grows as $\sqrt{x}$). At a critical distance x_{cr} the layer turns turbulent through a transition zone.
- Local Reynolds number $Re_x = U_\infty\, x / \nu$. Critical value for flat plate $Re_{cr} \approx 5 \times 10^5$ (range $3\times10^5 – 10^6$ depending on roughness & free-stream turbulence).
- In the turbulent layer a very thin **laminar sub-layer** persists next to the wall : $\delta' = 11.6\ \nu/u_*$ where shear velocity $u_* = \sqrt{(\tau_0/\rho)}$.
- Turbulent layer is thicker, fuller velocity profile, larger wall shear stress and resists separation better than laminar.
- Boundary layers can be laminar / turbulent over smooth or rough surfaces; for pipes the layer grows until it fills the pipe (entrance length $L \approx 0.06\ Re\ D$ laminar, ≈ 25–40 D turbulent).

Characteristics of boundary layer

Item	Laminar BL	Turbulent BL
Thickness (flat plate)	$\delta = 5x/\sqrt{Re_x}$ (Blasius)	$\delta = 0.37x/Re_x^{1/5}$
Velocity profile	parabolic / cubic	1/7 power law, fuller
Local C_f	$0.664/\sqrt{Re_x}$	$0.059/Re_x^{1/5}$
Average C_D	$1.328/\sqrt{Re_L}$	$0.074/Re_L^{1/5}$ ($5\times10^5 < Re_L < 10^7$)
Separation	early (poor resistance)	later (better resistance)

2. Boundary Layer Thicknesses

Because $u \to U$ asymptotically, δ is not sharply defined. More meaningful thicknesses are integral measures :



- **Displacement thickness δ^*** : distance by which the external streamlines are displaced outward due to the reduction of flow in the layer.

$$\delta^* = \int_0^{\delta} (1 - u/U)\, dy$$

- **Momentum thickness θ** : thickness of fluid layer which, moving at U, carries the momentum lost because of the boundary layer.

$$\theta = \int_0^{\delta} (u/U)(1 - u/U)\, dy$$

- **Energy thickness δ_e** : thickness of a layer at velocity U that carries the kinetic energy deficit.

$$\delta_e = \int_0^{\delta} (u/U)(1 - u^2/U^2)\, dy$$

- Shape factor $H = \delta^*/\theta$: ≈ 2.6 laminar (Blasius), 1.3–1.4 turbulent; H rises towards 3.5-4 near separation – a flag of impending separation.

Ex 1. For $u/U = 1.5(y/\delta) - 0.5(y/\delta)^3$: $\delta^* = \delta\int(1-1.5\eta+0.5\eta^3)d\eta = \delta(1 - 0.75 + 0.125) = 3\delta/8$; $\theta = 39\delta/280 = 0.139\delta$; $H = (3/8)/(39/280) = 2.69$.

3. Momentum Integral Equation (von Kármán)

Apply the momentum theorem to a control volume of length dx and height $h > \delta$ in the boundary layer (steady, 2-D, incompressible).

- Net momentum flux out in x-direction = $\rho \int u^2\, dy$ changes by $d/dx(\rho\int u^2 dy)dx$; mass is also entrained from the free stream carrying momentum U per unit mass.

- Force on the CV = $-\tau_0\,dx$ (wall shear) – (pressure term, via Bernoulli $dp/dx = -\rho U\,dU/dx$).

Result (valid for laminar & turbulent layers) :

$$\tau_0/\rho U^2 = d\theta/dx + (2\theta + \delta^*)/U \cdot dU/dx$$

For a flat plate (U = const, $dp/dx = 0$) :

$$\tau_0 = \rho U^2\, d\theta/dx \quad C_f = \tau_0/(\tfrac{1}{2}\rho U^2) = 2\, d\theta/dx$$

Application – laminar layer with cubic profile

Assume $u/U = (3/2)\eta - (1/2)\eta^3$, $\eta = y/\delta$.

- Wall shear : $\tau_0 = \mu\,(du/dy)_0 = 3\mu U/2\delta$.
- Momentum thickness $\theta = 39\delta/280 \Rightarrow d\theta/dx = (39/280)\,d\delta/dx$.
- Equate : $3\mu U/(2\delta) = \rho U^2(39/280)\,d\delta/dx \Rightarrow \delta\,d\delta = (140/13)(\nu/U)\,dx$. Integrate ($\delta = 0$ at $x = 0$) :

$$\delta = 4.64\,x/\sqrt{Re_x} \quad C_f = 0.646/\sqrt{Re_x} \quad C_D = 1.292/\sqrt{Re_L}$$

- Exact Blasius solution gives 5.0, 0.664 and 1.328 – so the approximate method is within 4 %.
- Drag on one side of a plate of length L, width b : $F_D = C_D \cdot \tfrac{1}{2}\rho U^2 \cdot (L\,b)$.

Ex 2. Air ($\nu = 1.5\times10^{-5}$ m²/s), U = 5 m/s : at x = 1 m, $Re_x = 3.33\times10^5$ (laminar) $\Rightarrow \delta = 5(1)/\sqrt{3.33\times10^5} = 8.7$ mm.

Ex 3. Smooth plate 2 m × 1 m in water ($\nu = 10^{-6}$) at U = 0.2 m/s : $Re_L = 4\times10^5$ (laminar) ; $C_D = 1.328/\sqrt{(4\times10^5)} = 0.0021$; F_D (one side) = 0.0021 × ½(1000)(0.2)² × 2 = 0.084 N.

Turbulent layer (1/7th power-law profile)

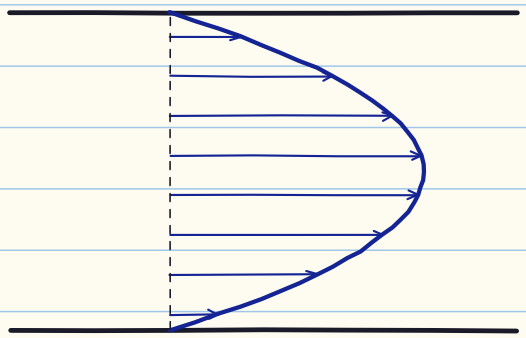
$u/U = (y/\delta)^{1/7} \Rightarrow \delta^* = \delta/8$, $\theta = 7\delta/72$. Using Blasius $\tau_0 = 0.0225\rho U^2(\nu/U\delta)^{1/4}$ in the momentum equation :

$$\delta = 0.37\,x\,Re_x^{-1/5} \quad C_f = 0.059\,Re_x^{-1/5} \quad C_D = 0.074\,Re_L^{-1/5}$$

- For plates laminar up to Re_{cr} then turbulent : $C_D = 0.074/Re_L^{1/5} - 1700/Re_L$ (Re_L up to 10^7).
- Rough surfaces : C_D depends on L/k_s – roughness raises drag in turbulent flow.

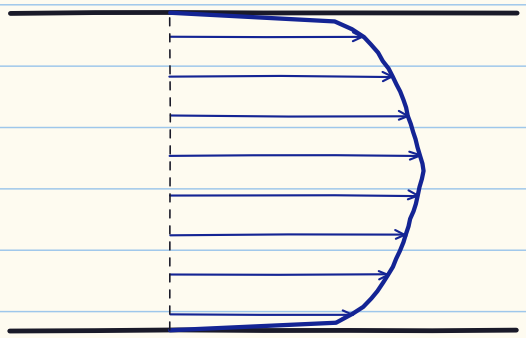
4. Velocity Profiles – Basic Concepts

LAMINAR : parabolic



$u_{max} = 2\ V_{avg}$

TURBULENT : flat (1/7 power law)



$V_{avg} \approx 0.8\ u_{max}$

- Laminar BL : $u/U = 2\eta - \eta^2$ (parabolic) or cubic (above) or sinusoidal $\sin(\pi\eta/2)$.
- Turbulent BL, wall region : universal log law (Prandtl–von Kármán)

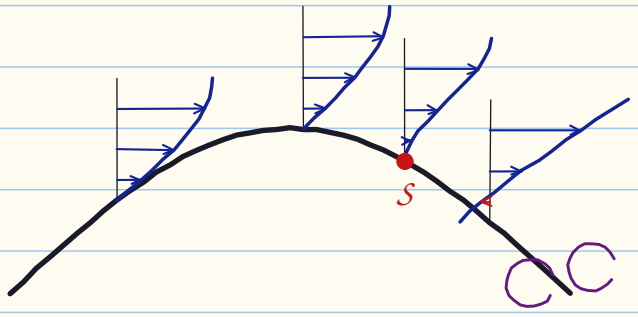
$$u/u_* = (1/\kappa)\ln(y\,u_*/\nu) + B \quad \kappa \approx 0.4\ ,\ B \approx 5.5$$

- Regions : viscous sub-layer ($y^+ < 5$, $u^+ = y^+$), buffer layer, log-law region, outer (wake) region.
- Flow in pipes : laminar parabolic $u = u_{max}[1 - (r/R)^2]$; turbulent $u/u_{max} = (y/R)^{1/n}$, $n \approx 7$ for $Re \approx 10^5$.

5. Separation of Boundary Layer

Consider flow over a curved surface (cylinder, aerofoil at high incidence, diffuser).

Flow →



Separation point S : $(\partial u/\partial y)_{y=0} = 0$

upstream of crest: $dp/dx < 0$ (favourable)

beyond crest: $dp/dx > 0$ (adverse)

reverse flow + wake ⇒ form drag

Control : suction, blowing, slots, streamlining, vortex generators

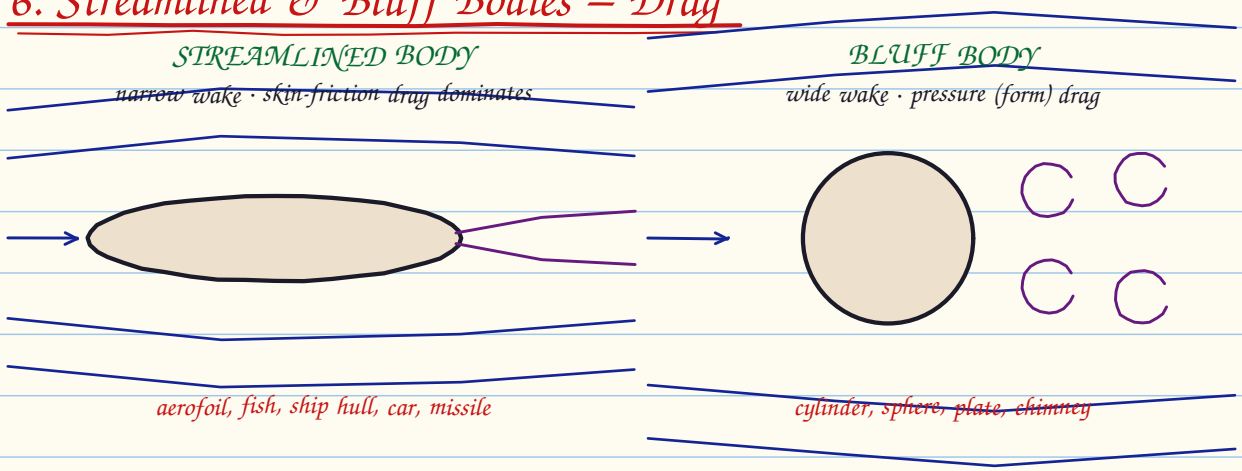
- Up to the crest the free-stream accelerates : $dp/dx < 0$ ⇒ favourable – layer thins, no separation.
- Beyond the crest the free stream decelerates : $dp/dx > 0$ ⇒ adverse pressure gradient. Fluid in the layer, already slowed by friction, loses KE and cannot climb the pressure hill.

- At the **point of separation S** : $(\partial u/\partial y)_{y=0} = 0$. Downstream, fluid near the wall flows backward → layer detaches, forms eddies/wake of low pressure.
- Consequences : large pressure (form) drag, loss of lift and stall of aerofoils, energy loss and noise, cavitation in machines, poor diffuser performance.
- Separation occurs only for adverse gradient (e.g. diverging flow); it cannot happen in converging flow. A turbulent layer separates later than laminar one – so golf balls have dimples and cricket balls are roughened to trip the layer.

Methods to control / delay separation

- Streamlining the shape (gentle taper, small divergence angle < 7° in diffusers).
- Suction of the slow layer through slots/porous walls (removes retarded fluid).
- Blowing / injection of high-energy fluid tangentially into the layer (flaps, slats on aircraft wings).
- Rotating (moving) wall in the direction of flow – Flettner rotor.
- Vortex generators / turbulators / trip wires – energise the layer to become turbulent.
- Guide vanes in bends and diffusers ; ensuring small wing angle of attack.

6. Streamlined & Bluff Bodies – Drag



Total drag = skin-friction drag (shear stress integrated over surface) + pressure (form) drag (pressure difference between front and wake).

$$F_D = C_D \cdot \tfrac{1}{2} \rho A U^2 \quad F_L = C_L \cdot \tfrac{1}{2} \rho A U^2$$

- A = projected area normal to flow (frontal) for bluff bodies ; plan area for plates/aerofoils.

- **Streamlined body** : tapered tail keeps the flow attached ; wake narrow ; drag mostly friction ; small C_D. Uses : aircraft, ships, submarines, vehicles, turbine blades, missiles.
- **Bluff body** : blunt shape, flow separates near the shoulder ; wide low-pressure wake ; drag mostly form drag. Uses : chimneys, bridge piers, buildings, poles, parachutes (large drag intended).

Body	C_D	Remark
Flat plate normal to flow	≈ 1.98	bluff
Circular cylinder ($10^3<Re<10^5$)	$\approx 1.0 – 1.2$	bluff
Sphere ($10^3<Re<2\times10^5$)	≈ 0.47	drops to 0.2 after critical Re
Streamlined strut / aerofoil	$0.04 – 0.1$	streamlined
Sphere, $Re < 1$ (Stokes)	$C_D = 24/Re$	$F_D = 3\pi\mu\, d\, U$

- Karman vortex street : alternate vortices shed behind a cylinder, Strouhal number $St = f\, d/U \approx 0.2$; may cause resonance (chimneys, cables, bridges).
- Terminal velocity : weight − buoyancy = drag ⇒ Stokes : $V_t = d^2(\rho_s - \rho)g/(18\mu)$.

Ex 4. Chimney 40 m high, 2 m dia, wind 25 m/s ($\rho = 1.2$), $C_D = 0.7$: $F_D = 0.7 \times \tfrac{1}{2}(1.2)(40\times2)(25)^2 = 21$ kN.

7. Dimensional Analysis

Method of reducing the number of variables in a physical problem by combining them into dimensionless groups.

- **Dimensional homogeneity** (Fourier) : every term of a physically correct equation has the same dimensions. e.g. Bernoulli : $p/\rho g$, $v^2/2g$, z all have dimension L.
- Uses : (i) check equations ; (ii) derive relations among variables ; (iii) plan experiments with fewer variables ; (iv) model testing – scale-up to prototype.
- Methods : Rayleigh's indicial method (≤ 4 variables) and Buckingham π-theorem (general).

Non-dimensionalisation of equations

Choose reference scales L (length), U (velocity), ρU^2 (pressure). Let $x^* = x/L$, $u^* = u/U$, $p^* = p/\rho U^2$. Navier–Stokes for incompressible flow becomes :

$$\partial u^*/\partial t^* + u^*\cdot\nabla u^* = -\nabla p^* + (1/Re)\,\nabla^2 u^* + (1/Fr^2)\,\hat{g}$$

- Each dimensionless group emerges naturally : $Re = \rho UL/\mu$, $Fr = U/\sqrt{(gL)}$ ⇒ two flows are dynamically similar if these groups are equal.

Buckingham π-theorem

If a physical phenomenon involves n variables and these contain m fundamental dimensions (M, L, T), the relation can be written as a function of (n − m) independent dimensionless π-terms : $f(\pi_1, \pi_2, \dots \pi_{n-m}) = 0$.

Method of repeating variables – steps

- 1. List all n variables (include dimensional constants such as g if needed).
- 2. Write dimensions of each in M, L, T ; find m.
- 3. Choose m repeating variables : together contain all m dimensions, no one dimensionless, no two with the same dimensions. Usual choice : one geometric (D or L), one kinematic (V), one dynamic (ρ) – never the dependent variable.
- 4. Form π-terms : each π = (non-repeating variable) × (repeating)^a (repeating)^b (repeating)^c ; solve exponents so π is dimensionless.
- 5. Express result $\pi_1 = \varphi(\pi_2, \pi_3 \dots)$; replace any π by a power or product of itself if needed (e.g. 1/Re).

Ex 5 (drag on a sphere). $F = f(D, V, \rho, \mu)$. $n = 5$, $m = 3$ ⇒ 2 π-terms. Repeating : D, V, ρ. $\pi_1 = F D^a V^b \rho^c$: M : $1 + c = 0$; T : $-2 - b = 0$; L : $1 + a + b - 3c = 0$ ⇒ $c = -1$, $b = -2$, $a = -2$ ⇒ $\pi_1 = F/(\rho V^2 D^2)$. $\pi_2 = \mu/(\rho V D) = 1/Re$. $F = \rho V^2 D^2 \varphi(Re)$ ⇒ $C_D = \varphi(Re)$.

Ex 6 (pipe pressure drop). $\Delta p = f(D, L, V, \rho, \mu, \varepsilon)$. $n = 7$, $m = 3$ ⇒ 4 π-terms. With D, V, ρ : $\Delta p/(\rho V^2) = \varphi(L/D, \varepsilon/D, Re)$; since $\Delta p \propto L$: $\Delta p = \rho V^2 (L/D) \varphi_1(Re, \varepsilon/D)$ ⇒ $h_f = f L V^2/2gD$ (Darcy-Weisbach recovered, $f = \varphi_1$).

Important dimensionless numbers

Number	Ratio	Use
Reynolds $Re = \rho VL/\mu$	inertia / viscous	pipe flow, submarines, aircraft (no free surface)
Froude $Fr = V/\sqrt{(gL)}$	inertia / gravity	open channels, ships, weirs
Euler $Eu = \Delta p/\rho V^2$	pressure / inertia	pumps, cavitation (σ)

Number	Ratio	Use
Weber $We = \rho V^2 L/\sigma$	inertia / surface tension	capillary flow, droplets
Mach $Ma = V/c$	flow / sound speed	compressible flow
Strouhal $St = fL/V$	unsteady / convective	vortex shedding

Model similarity

- **Geometric** : same shape, constant scale ratio $L_r = L_p/L_m$. **Kinematic** : velocity ratios same at corresponding points (same streamline pattern). **Dynamic** : force ratios equal (same Re, Fr…).
- Reynolds model law : $V_r = \nu_r/L_r$. Froude model law : $V_r = \sqrt{L_r}$, $Q_r = L_r^{5/2}$.

Ex 7. A ship model 1:25 is towed at 1.2 m/s. Prototype speed (Froude) : $V_p = V_m\sqrt{25} = 6$ m/s ; prototype wave drag = model drag × 25^3.

8. Lift, Stall & Magnus Effect (short notes)

- **Lift** on an aerofoil arises because the faster flow over the upper surface has lower pressure than the lower surface (Bernoulli) – equivalently, the circulation Γ gives $F_L = \rho\, U\, \Gamma\, b$ (Kutta–Joukowski).
- **Stall** : at large angle of attack ($\approx 15°$) the layer separates from the upper surface, C_L falls sharply, C_D rises. Slats, slots and flaps delay stall by re-energising the layer.
- **Magnus effect** : a spinning cylinder/sphere in a stream experiences a transverse force (circulation superposed on uniform flow) – swing of a cricket/football, Flettner rotor ship.
- **Induced drag** arises from wing-tip vortices ; winglets reduce it.
- Pipe entrance : boundary layers from the wall merge at the entrance length ; beyond this the profile is fully developed.

9. Additional Solved Problems

P1 (laminar plate). Water ($\nu = 10^{-6}$), $U = 0.4$ m/s, plate length 1 m : $Re_L = 4\times10^5$ (laminar). $\delta_L = 5(1)/\sqrt{(4\times10^5)} = 7.9$ mm ; C_f at end $= 0.664/632.5 = 0.00105 \Rightarrow \tau_0 = 0.00105 \times \tfrac{1}{2}(1000)(0.4)^2 = 0.084$ Pa ; $C_D = 1.328/632.5 = 0.0021 \Rightarrow F_D$ (one side, per metre width) $= 0.0021 \times 80 \times 1 = 0.168$ N.

P2 (turbulent plate). Air ($\rho = 1.2$, $\nu = 1.5\times10^{-5}$), $U = 30$ m/s, $L = 3$ m, $b = 1$ m : $Re_L = 6\times10^6$ (mixed layer). $C_D = 0.074/(6\times10^6)^{0.2} - 1700/6\times10^6 = 0.00326 - 0.00028 = 0.00298$. $F_D = 0.00298 \times \tfrac{1}{2}(1.2)(30)^2 \times 3 = 4.8$ N (one side).

P3 (Rayleigh's method). Time period of a simple pendulum $T = f(l, g)$. $T = K\, l^a\, g^b \Rightarrow [T] : 1 = -2b \Rightarrow b = -\tfrac{1}{2}$; $[L] : 0 = a + b \Rightarrow a = \tfrac{1}{2} \Rightarrow T = K\sqrt{(l/g)}$.

P4 (π-theorem for a pump / fan). Power $P = f(\rho, N, D)$ with $m = 3$, $n = 4 \Rightarrow 1$ π-term : $\pi = P\,\rho^a N^b D^c$; $M : 1 + a = 0$; $T : -3 - b = 0$; $L : 2 - 3a + c = 0 \Rightarrow a = -1$, $b = -3$, $c = -5 \Rightarrow P = K\rho\, N^3\, D^5$ (power coefficient).

P5 (Stokes' law). Sphere $d = 1$ mm, $\rho_s = 2650$, falling in oil ($\rho = 900$, $\mu = 0.1$ Pa·s) : $V_t = d^2(\rho_s - \rho)g/(18\mu) = 10^{-6}(1750)(9.81)/(1.8) = 9.5$ mm/s ; check $Re = 900(0.00954)(0.001)/0.1 = 0.086 < 1$ ✓ (Stokes valid).

P6 (Froude model). Spillway model scale 1 : 20, model discharge 0.02 m^3/s : $Q_r = L_r^{5/2} = 20^{2.5} = 1789 \Rightarrow Q_p = 0.02 \times 1789 = 35.8\ m^3/s$.

Common mistakes to avoid

- Taking δ as the point where $u = U$ exactly – use 0.99 U. Using laminar formulas when $Re_L > 5\times10^5$.
- Selecting repeating variables that are not independent (e.g. V, L and Q) or including the dependent variable among them.
- Forgetting that separation needs an *adverse* gradient – accelerating flow never separates.
- Mixing up area : bluff body → frontal area ; plate/aerofoil → plan area.

10. Short Answers – Quick Revision

- **Why does the boundary layer thicken downstream?** Viscous effect diffuses further from the wall as more fluid is slowed ; $\delta \propto \sqrt{x}$ (laminar).
- **Why is wall shear greater in a turbulent layer?** Fuller profile gives a steeper gradient $(du/dy)_{wall}$; momentum exchange by eddies.

- *Why use dimples on a golf ball?* They trip the layer to turbulence, delaying separation, shrinking the wake and cutting form drag by ~ 50 %.
- *What is d'Alembert's paradox?* Potential flow past a body predicts zero drag – resolved by the boundary layer & wake.
- *Is displacement thickness always smaller than δ?* Yes ; $\delta^* < \theta \cdot H$ and $\delta^* < \delta$ (for turbulent $\delta^* \approx \delta/8$, laminar $\approx 0.375\delta$).
- *What if two numbers (Re and Fr) must be matched?* Usually impossible with same fluid – choose the dominant force (Fr for free-surface, Re for pressure-driven flows).
- *Number of π-terms for 6 variables, 3 dimensions?* $6 - 3 = 3$.
- *When is the 1/7th power law invalid?* $Re > 10^7$ (use log law), near the wall ($y^+ < 30$) and in separated flow.

11. Quick Formula Sheet – Unit III

Item	Formula
Thicknesses	$\delta^* = \int(1-u/U)dy$; $\theta = \int(u/U)(1-u/U)dy$
Momentum integral	$\tau_0 = \rho U^2 \, d\theta/dx$
Laminar flat plate	$\delta = 5x/\sqrt{Re_x}$; $C_D = 1.328/\sqrt{Re_L}$
Turbulent flat plate	$\delta = 0.37x/Re_x^{0.2}$; $C_D = 0.074/Re_L^{0.2}$
Drag, lift	$F = C \cdot \frac{1}{2}\rho A U^2$
Separation	$(\partial u/\partial y)_{y=0} = 0$; $dp/dx > 0$
π-theorem	number of $\pi = n - m$
Froude / Reynolds model	$V_r = \sqrt{L_r}$; $V_r = \nu_r/L_r$

12. Important Questions

- Define boundary layer. Explain its growth over a flat plate with a neat sketch; laminar sub-layer.
- Define displacement, momentum and energy thickness and derive expressions. Find H for a given profile.
- Derive von Kármán momentum integral equation. Use it to obtain δ, C_f, C_D for a laminar layer with cubic profile.

- What is separation? Explain with velocity profiles; state methods of control of separation.
- Distinguish streamlined and bluff bodies with examples; define drag & lift coefficients.
- State and prove Buckingham π-theorem steps; solve a problem by repeating-variable method. State uses of dimensionless numbers.